

Cloud Monitoring: Qwik Start - LAB

GSP089



Overview

Cloud Monitoring provides visibility into the performance, uptime, and overall health of cloud-powered applications. Cloud Monitoring collects metrics, events, and metadata from Google Cloud, Amazon Web Services, hosted uptime probes, application instrumentation, and a variety of common application components including Cassandra, Nginx, Apache Web Server, Elasticsearch, and many others. Cloud Monitoring ingests that data and generates insights via dashboards, charts, and alerts. Cloud Monitoring alerting helps you collaborate by integrating with Slack, PagerDuty, HipChat, Campfire, and more.

In this lab you'll install monitoring and logging agents to collect information from your instance, which could include metrics and logs from 3rd party apps.

Objectives

In this lab, you learn how to:

- Monitor a Compute Engine virtual machine (VM) instance with Cloud Monitoring.
- Install monitoring and logging agents for your VM

Q

Dashboard

Catalog

Paths

Collections

Home > Develop Your Google Cloud Network > Cloud Monitoring: Qwik Start

Contents

Cloud Monitoring: Qwik Start

Lab setup instructions and requirements

End Lab 00:47:52

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username

student-02-44c3c12d53fd0

Password

MEB8TZKXavPu

Project ID

qwiklabs-gcp-01-0bae0d0f

gcloud is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

3. (Optional) You can list the active account name with this command:

gcloud auth list

4. Click **Authorize**.

Output:

ACTIVE: *
ACCOUNT: student-02-44c3c12d53fd@qwiklabs.net

To set the active account, run:
\$ gcloud config set account `ACCOUNT`

5. (Optional) You can list the project ID with this command:

Lab instructions and tasks

0/100

GSP089

Overview

Objectives

Setup and requirements

Task 1. Create a Compute Engine instance

Task 2. Add Apache2 HTTP Server to your instance

Task 3. Create an uptime check

Task 4. Create an alerting policy

Task 5. Create a



[Develop Your Google Cloud Network](#) > [Cloud Monitoring: Qwik Start](#)

— Contents —

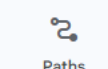
Cloud Monitoring: Qwik Start



Dashboard



Catalog



Paths



Collections

Lab setup instructions and requirements

End Lab

00:47:29

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked.
[Learn more.](#)

[Open Google Cloud console](#)

Username

student-02-44c3c12d53fd0



Password

MEB8TZKXavPu



Project ID

qwiklabs-qcp-01-0bae0d00



Set your region and zone

Certain Compute Engine resources live in regions and zones. A region is a specific geographical location where you can run your resources. Each region has one or more zones.

Note: Learn more about regions and zones and see a complete list in [Regions & Zones documentation](#).

Run the following `gcloud` commands in Cloud Shell to set the default region and zone for your lab:

```
gcloud config set compute/zone "us-east1-c"
export ZONE=$(gcloud config get compute/zone)

gcloud config set compute/region "us-east1"
export REGION=$(gcloud config get compute/region)
```



Lab instructions and tasks

0/100

GSP089

Overview

Objectives

Setup and requirements

Task 1. Create a Compute Engine instance

Task 2. Add Apache2 HTTP Server to your instance

Task 3. Create an uptime check

Task 4. Create an alerting policy

Task 5. Create a

[← Previous](#)

Next >

Dashboard – qwiklabs-gcp-01-01

qwiklabs-gcp-01-0ba... – Google Clo

+

< > ↺ 🏠

🔖

🔗

https://console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-01-0bae0d009b16&pli=1&cloudshell=true

🔍

📅

🖨

🗨

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰

Google Cloud

🔗 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦

📄

🔔

?

⋮

S

Dashboard

Activity

Recommendations

✎ Customize

📄

CLOUD SHELL

Terminal

qwiklabs-gcp-01-0bae0d009b16

✕

+

▼

✎ Open Editor

📄

⚙

📷

📺

⋮

–

↕

🗨

✕

📘

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)

Don't show again

Dismiss

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to **qwiklabs-gcp-01-0bae0d009b16**.
Use ``gcloud config set project [PROJECT_ID]`` to change to a different project.
student_02_44c3c12d53fd@cloudshell:~ (**qwiklabs-gcp-01-0bae0d009b16**)\$ `gcloud auth list`
Credentialed Accounts

ACTIVE: *
ACCOUNT: student-02-44c3c12d53fd@qwiklabs.net

To set the active account, run:
\$ `gcloud config set account `ACCOUNT``

student_02_44c3c12d53fd@cloudshell:~ (**qwiklabs-gcp-01-0bae0d009b16**)\$ `gcloud config list project`
INFORMATION: Project 'qwiklabs-gcp-01-0bae0d009b16' has no 'environment' tag set. Use either 'Production', 'Development', 'Test', or 'Staging'. Add an 'environment' tag using ``gcloud resource -manager tags bindings create``.
[core]
project = qwiklabs-gcp-01-0bae0d009b16

Your active configuration is: [cloudshell-20847]
student_02_44c3c12d53fd@cloudshell:~ (**qwiklabs-gcp-01-0bae0d009b16**)\$ `gcloud config set compute/zone "us-east1-c"`
export ZONE=\$(gcloud config get compute/zone)

`gcloud config set compute/region "us-east1"`
export REGION=\$(gcloud config get compute/region)
Updated property [compute/zone].
Your active configuration is: [cloudshell-20847]
Updated property [compute/region].
Your active configuration is: [cloudshell-20847]
student_02_44c3c12d53fd@cloudshell:~ (**qwiklabs-gcp-01-0bae0d009b16**)\$

9+ Air: Very poor Now

🪟 🔍 Search 🖱

📁 🗨 📄 📅 📁 📄 📄

⬆

ENG IN

📶 🔊 🔌

07:26 PM 27-11-2025

🔔

Q

Dashboard

Catalog

Paths

Collections

Home > Develop Your Google Cloud Network > Cloud Monitoring: Qwik Start

Contents

Cloud Monitoring: Qwik Start

Lab setup instructions and requirements

End Lab 00:47:09

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username

student-02-44c3c12d53fd0

Password

MEB8TZKXavPu

Project ID

qwiklabs-gcp-01-0bae0d0f

```
export REGION=$(gcloud config get compute/region)
```

Task 1. Create a Compute Engine instance

- In the **Cloud console**, on the **Navigation menu** (≡), click **Compute Engine > VM Instances**, then click **Create instance**.

Fill in the fields as follows, leaving all other fields at the default value:
- In the **Machine configuration**

Enter the values for the following fields:

Field	Value
Name	lamp-1-vm
Region	us-east1
Zone	us-east1-c

Lab instructions and tasks

0/100

GSP089

Overview

Objectives

Setup and requirements

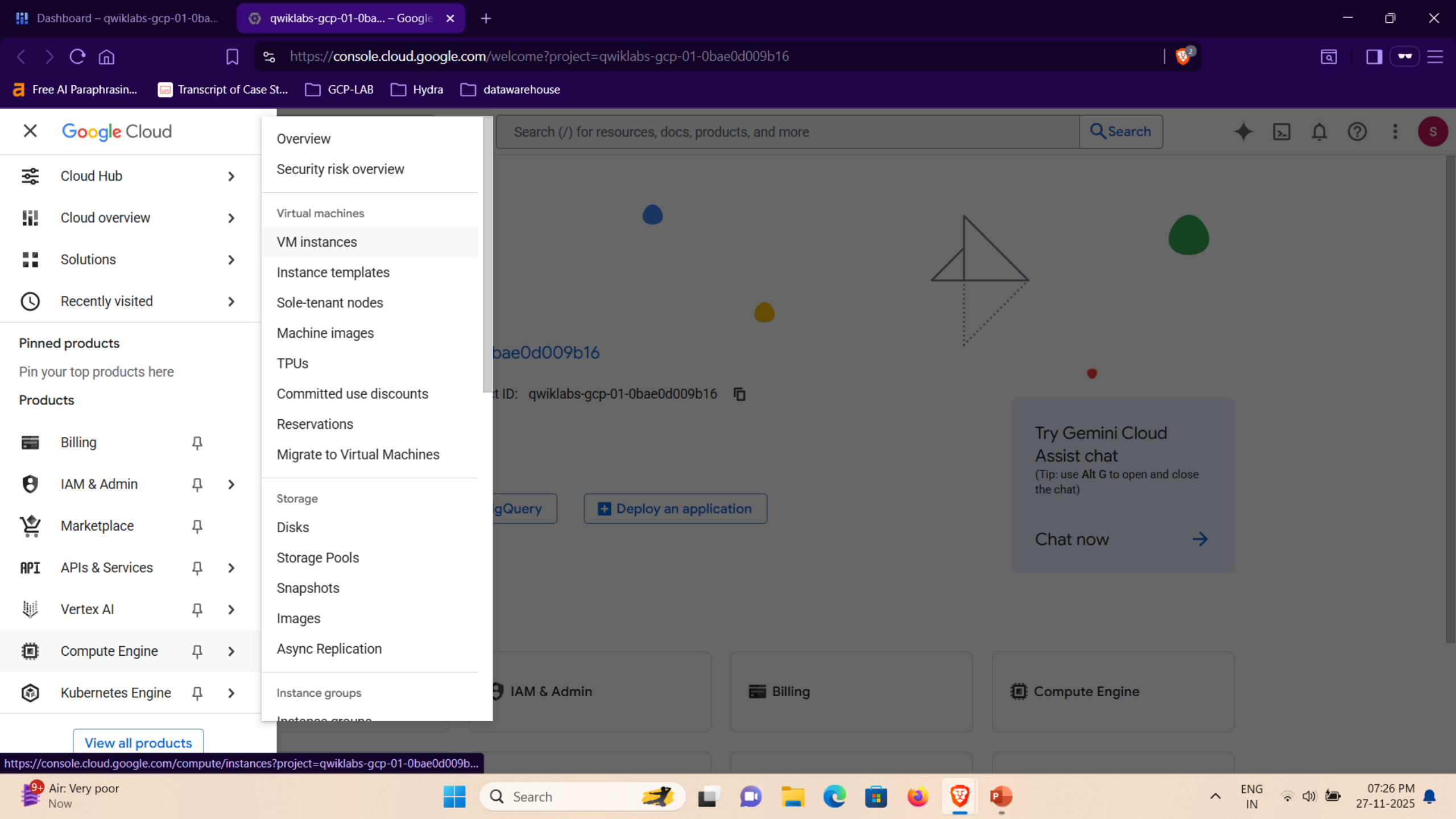
Task 1. Create a Compute Engine instance

Task 2. Add Apache2 HTTP Server to your instance

Task 3. Create an uptime check

Task 4. Create an alerting policy

Task 5. Create a



Overview

Security risk overview

Virtual machines

VM instances

Instance templates

Sole-tenant nodes

Machine images

TPUs

Committed use discounts

Reservations

Migrate to Virtual Machines

Storage

Disks

Storage Pools

Snapshots

Images

Async Replication

Instance groups

Instance groups

Search (/) for resources, docs, products, and more

Search

bae0d009b16

Project ID: qwiklabs-gcp-01-0bae0d009b16

gQuery

Deploy an application

Try Gemini Cloud

Assist chat

(Tip: use Alt G to open and close the chat)

Chat now



IAM & Admin

Billing

Compute Engine

View all products

https://console.cloud.google.com/compute/instances?project=qwiklabs-gcp-01-0bae0d009b16

9+ Air: Very poor
Now



Search



ENG
IN



07:26 PM
27-11-2025



←

Create an instance

⚡ Create VM from...

↔ Equivalent code

⏪

• Machine configuration
e2-medium, us-east1-c

• OS and storage
Debian GNU/Linux 12 (bookworm)

• Data protection
Snapshot schedules

• Networking
1 network interface

• Observability
Install Ops Agent

• Security

• Advanced

Machine configuration

Name *

lamp-1-vm

?

Region *

us-east1 (South Carolina)

▼

?

Region is permanent

Zone *

us-east1-c

▼

?

Zone is permanent

💡 NEW: General-purpose AMD machine series now available

< 1 of 2 > ✕

▼ Try the new N4D series, a high performance and cost-optimized machine series ideal

Try now

✓ General purpose

Compute optimized

Memory optimized

Storage optimized

GPUs

Machine types for common workloads, optimized for cost and flexibility

	Series ?	Description	vCPUs ?	Memory ?	CPU Platfor
☐	C4	Consistently high performance	2 - 288	4 - 2,232 GB	Intel Emerald
☐	C4A	Arm-based consistently high performance	1 - 72	2 - 576 GB	Google Axion
☐	C4D	Consistently high performance	2 - 384	3 - 3,072 GB	AMD Turin
☐	N4	Flexible & cost-optimized	2 - 80	4 - 640 GB	Intel Emerald

Create

Cancel

↔ Equivalent code

Monthly estimate

\$25.46

That's about \$0.03 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$24.46
10 GB balanced persistent disk	\$1.00
Logging	Cost varies
Monitoring	Cost varies
Snapshot schedule	Cost varies
Total	\$25.46

[Compute Engine pricing](#)

[Cloud Operations pricing](#)

^ Less

Dashboard – qwiklabs-gcp-01-0ba...

Compute Engine – qwiklabs-gcp

< > ↺ 🏠

🔖

🌐

https://console.cloud.google.com/compute/instancesAdd?project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅

🖨

👤

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰

Google Cloud

🔗 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦

📄

🔔

?

⋮

S

←

Create an instance

🌟 Create VM from...

↔️ Equivalent code

⏪

• Machine configuration
e2-medium, us-east1-c

• OS and storage
Debian GNU/Linux 12 (bookworm)

• Data protection
Snapshot schedules

• **Networking**
1 firewall rule, 1 network interface

• Observability
Install Ops Agent

• Security

• Advanced

Networking

Firewall ?
Select firewall rules to allow specific network traffic from the Internet. Firewall rules will only be applied when an instance is created.
☒ Allow HTTP traffic
☐ Allow HTTPS traffic
☐ Allow Load Balancer Health Checks

Network tags
Network tags
http-server X ?

Hostname
Hostname ?
Set a custom hostname for this instance or leave it default. Choice is permanent

IP forwarding ?
☐ Enable

Network performance configuration
Network bandwidth ?

Monthly estimate

\$25.46

That's about \$0.03 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$24.46
10 GB balanced persistent disk	\$1.00
Logging	Cost varies
Monitoring	Cost varies
Snapshot schedule	Cost varies
Total	\$25.46

[Compute Engine pricing](#)
[Cloud Operations pricing](#)
[^ Less](#)

Create

Cancel

↔️ Equivalent code

9+ Air: Very poor Now

🪟 🔍 Search 🖱 📁 🗂 🌐 📄 📅 📧

ENG IN 📶 🔊 🔋 07:28 PM 27-11-2025 🔔

Dashboard – qwiklabs-gcp-01-0ba...

VM instances – Compute Engine

< > ↺ 🏠

🔖

🌐 https://console.cloud.google.com/compute/instances?onCreate=true&project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅

🖨

👤

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰

Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦

📄

3

?

⋮

S

🔧 Compute Engine

📄 Overview

🛡 Security risk overview

Virtual machines

📄 VM instances

📄 Instance templates

👤 Sole-tenant nodes

📄 Machine images

🖨 TPUs

📊 Committed use discou...

📅 Reservations

🔄 Migrate to Virtual Mach...

Storage

📄 Disks

🛒 Marketplace

📄 Release Notes

<|

VM instances

Create instance

Import VM

Refresh

Learn

Instances

Observability

Instance schedules

VM instances

Filter Enter property name or value

☐	Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect	
☐	✓	lamp-1-vm	us-east1-c			10.142.0.2 (nic0)	104.196.99.84 (nic0)	SSH	⋮

Related actions

Show

🇵🇰 9+ PAK - SL

Live

🪟

🔍 Search

👤

📁

💬

📁

🌐

🛒

🔥

🛡

📄

⬆

ENG IN

📶

🔊

🔋

07:29 PM

27-11-2025

🔔

Q

Dashboard

Catalog

Paths

Collections

Home > Develop Your Google Cloud Network > Cloud Monitoring: Qwik Start

Contents

Cloud Monitoring: Qwik Start

Lab setup instructions and requirements >

End Lab 00:43:33

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username

student-02-44c3c12d53fd0

Password

MEB8TZKXavPu

Project ID

qwiklabs-gcp-01-0bae0d0f

✓

Create a Compute Engine instance

Check my progress

Assessment Completed!

Task 2. Add Apache2 HTTP Server to your instance

1. In the Console, click **SSH** in line with `lamp-1-vm` to open a terminal to your instance.

2. Run the following commands in the SSH window to set up Apache2 HTTP Server:

sudo apt-get update

Lab instructions and tasks

0/100

GSP089

Overview

Objectives

Setup and requirements

Task 1. Create a Compute Engine instance

Task 2. Add Apache2 HTTP Server to your instance

Task 3. Create an uptime check

Task 4. Create an alerting policy

Task 5. Create a

ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-01-0bae0d009b16/zones/us-east1-c/instances/lamp-1-vm?authuser=0&hl=en_US&projectNumber=655743958698&useAdminProxy=true

SSH-in-browser

UPLOAD FILE

DOWNLOAD FILE

Linux lamp-1-vm 6.1.0-40-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1 (2025-09-20) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.

Creating directory '/home/student-02-44c3c12d53fd'.

student-02-44c3c12d53fd@lamp-1-vm:~\$ sudo apt-get update

Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:2 file:/etc/apt/mirrors/debian-security.list Mirrorlist [39 B]
Hit:7 https://packages.cloud.google.com/apt google-compute-engine-bookworm-stable InRelease
Hit:3 https://deb.debian.org/debian bookworm InRelease
Hit:4 https://deb.debian.org/debian bookworm-updates InRelease
Hit:5 https://deb.debian.org/debian bookworm-backports InRelease
Hit:6 https://deb.debian.org/debian-security bookworm-security InRelease
Hit:8 https://packages.cloud.google.com/apt cloud-sdk-bookworm InRelease
Hit:9 https://packages.cloud.google.com/apt google-cloud-ops-agent-bookworm-2 InRelease
Reading package lists... Done

student-02-44c3c12d53fd@lamp-1-vm:~\$

PAK - SL
Live

Search

ENG
IN

07:30 PM
27-11-2025

Dashboard

Catalog

Paths

Collections

Cloud Monitoring: Qwik Start

Contents

Cloud Monitoring: Qwik Start

Lab setup instructions and requirements

End Lab 00:40:51

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username student-02-44c3c12d53fd

Password MEB8TZKXavPu

Project ID qwiklabs-gcp-01-0bae0d0f

3. When asked if you want to continue, enter Y.

Note: If you cannot install php7.0, use php5.

sudo service apache2 restart

Click **Check my progress** below. A green check confirms you're on track.

✓

Check my progress

Assessment Completed!

4. Return to the Cloud Console, on the VM instances page. Click the **External IP** for **lamp-1-vm** instance to see the Apache2 default page for this instance.

Note: If you are unable to find the **External IP** column then click on **Column Display**

25/100

Lab instructions and tasks

GSP089

Overview

Objectives

Setup and requirements

Task 1. Create a Compute Engine instance

Task 2. Add Apache2 HTTP Server to your instance

Task 3. Create an uptime check

Task 4. Create an alerting policy

Task 5. Create a

< Previous

Next >

ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-01-0bae0d009b16/zones/us-east1-c/instances/lamp-1-vm?authuser=0&hl=en_US&projectNumber=655743958698&useAdminProxy=true

SSH-in-browser

UPLOAD FILE

DOWNLOAD FILE

```
student-02-44c3c12d53fd@lamp-1-vm:~$ sudo apt-get install apache2 php7.0
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'php7.0-thrift' for regex 'php7.0'
Note, selecting 'php7.0-common' for regex 'php7.0'
Note, selecting 'php7.0-curl' for regex 'php7.0'
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libgdbm-compat4 libjansson4 liblua5.3-0 libperl5.36 perl perl-modules-5.36 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser perl-doc libterm-readline-gnu-perl | libterm-readline-perl-perl make libtap-harness-archive-perl
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libgdbm-compat4 libjansson4 liblua5.3-0 libperl5.36 perl perl-modules-5.36
  ssl-cert
0 upgraded, 15 newly installed, 0 to remove and 4 not upgraded.
Need to get 9698 kB of archives.
After this operation, 57.1 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:2 https://deb.debian.org/debian bookworm/main amd64 perl-modules-5.36 all 5.36.0-7+deb12u3 [2815 kB]
Get:3 https://deb.debian.org/debian bookworm/main amd64 libgdbm-compat4 amd64 1.23-3 [48.2 kB]
Get:4 https://deb.debian.org/debian bookworm/main amd64 libperl5.36 amd64 5.36.0-7+deb12u3 [4196 kB]
Get:5 https://deb.debian.org/debian bookworm/main amd64 perl amd64 5.36.0-7+deb12u3 [239 kB]
Get:6 https://deb.debian.org/debian bookworm/main amd64 libapr1 amd64 1.7.2-3+deb12u1 [102 kB]
Get:7 https://deb.debian.org/debian bookworm/main amd64 libaprutil1 amd64 1.6.3-1 [87.8 kB]
Get:8 https://deb.debian.org/debian bookworm/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1 [13.6 kB]
Get:9 https://deb.debian.org/debian bookworm/main amd64 libaprutil1-ldap amd64 1.6.3-1 [11.8 kB]
Get:10 https://deb.debian.org/debian bookworm/main amd64 libjansson4 amd64 2.14-2 [40.8 kB]
Get:11 https://deb.debian.org/debian bookworm/main amd64 liblua5.3-0 amd64 5.3.6-2 [123 kB]
Get:12 https://deb.debian.org/debian bookworm/main amd64 apache2-bin amd64 2.4.65-1~deb12u1 [1396 kB]
Get:13 https://deb.debian.org/debian bookworm/main amd64 apache2-data all 2.4.65-1~deb12u1 [160 kB]
Get:14 https://deb.debian.org/debian bookworm/main amd64 apache2-utils amd64 2.4.65-1~deb12u1 [214 kB]
Get:15 https://deb.debian.org/debian bookworm/main amd64 apache2 amd64 2.4.65-1~deb12u1 [228 kB]
Get:16 https://deb.debian.org/debian bookworm/main amd64 ssl-cert all 1.1.2 [21.1 kB]
Fetched 9698 kB in 0s (32.2 MB/s)
Preconfiguring packages ...
Selecting previously unselected package perl-modules-5.36.
(Reading database ... 73802 files and directories currently installed.)
Preparing to unpack .../00-perl-modules-5.36_5.36.0-7+deb12u3_all.deb ...
Unpacking perl-modules-5.36 (5.36.0-7+deb12u3) ...
Selecting previously unselected package libgdbm-compat4:amd64.
Preparing to unpack .../01-libgdbm-compat4_1.23-3_amd64.deb ...
Unpacking libgdbm-compat4:amd64 (1.23-3) ...
Selecting previously unselected package libperl5.36:amd64.
Preparing to unpack .../02-libperl5.36_5.36.0-7+deb12u3_amd64.deb ...
Unpacking libperl5.36:amd64 (5.36.0-7+deb12u3) ...
```

9+ 16°C Clear

Search

ENG IN

07:32 PM 27-11-2025

ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-01-0bae0d009b16/zones/us-east1-c/instances/lamp-1-vm?authuser=0&hl=en_US&projectNumber=655743958698&useAdminProxy=true

SSH-in-browser

UPLOAD FILE

DOWNLOAD FILE

Setting up libapr1:amd64 (1.7.2-3+deb12u1) ...
Setting up libjansson4:amd64 (2.14-2) ...
Setting up perl-modules-5.36 (5.36.0-7+deb12u3) ...
Setting up ssl-cert (1.1.2) ...
Setting up libgdbm-compat4:amd64 (1.23-3) ...
Setting up liblua5.3-0:amd64 (5.3.6-2) ...
Setting up apache2-data (2.4.65-1~deb12u1) ...
Setting up libperl5.36:amd64 (5.36.0-7+deb12u3) ...
Setting up libaprutil1:amd64 (1.6.3-1) ...
Setting up libaprutil1-ldap:amd64 (1.6.3-1) ...
Setting up libaprutil1-dbd-sqlite3:amd64 (1.6.3-1) ...
Setting up perl (5.36.0-7+deb12u3) ...
Setting up apache2-utils (2.4.65-1~deb12u1) ...
Setting up apache2-bin (2.4.65-1~deb12u1) ...
Setting up apache2 (2.4.65-1~deb12u1) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /lib/systemd/system/apache-htcacheclean.service.
Processing triggers for man-db (2.11.2-2) ...
Processing triggers for libc-bin (2.36-9+deb12u13) ...
student-02-44c3c12d53fd@lamp-1-vm:~\$ sudo service apache2 restart
student-02-44c3c12d53fd@lamp-1-vm:~\$

9+ 16°C
Clear

Search

ENG
IN

07:32 PM
27-11-2025

Search

Dashboard

Catalog

Paths

Collections

Home > Develop Your Google Cloud Network > Cloud Monitoring: Qwik Start

Contents

Cloud Monitoring: Qwik Start

Lab setup instructions and requirements

End Lab 00:39:49

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username student-02-44c3c12d53fd0

Password MEB8TZKXavPu

Project ID qwiklabs-gcp-01-0bae0d0f

Assessment Completed!

4. Return to the Cloud Console, on the VM instances page. Click the External IP for lamp-1-vm instance to see the Apache2 default page for this instance.

Note: If you are unable to find External IP column then click on Column Display Options icon on the right side of the corner, select External IP checkbox and click OK.

Click **Check my progress** below. A green check confirms you're on track.

Get a success response over External IP of VM instance

Check my progress

Assessment Completed!

Create a Monitoring Metrics Scope

Lab instructions and tasks

75/100

GSP089

Overview

Objectives

Setup and requirements

Task 1. Create a Compute Engine instance

Task 2. Add Apache2 HTTP Server to your instance

Task 3. Create an uptime check

Task 4. Create an alerting policy

Task 5. Create a

Dashboard – qwiklabs-gcp-01-0ba...

VM instances – Compute Engine

Apache2 Debian Default Page: It worl

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/compute/instances?onCreate=true&project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅

📄

👤 2

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰

Google Cloud

🔍 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦

📄

🟢 3

?

⋮

🟠 S

🔧 Compute Engine

📌

VM instances

Create instance

Import VM

Refresh

Learn

📊 Overview

🛡️ Security risk overview

Virtual machines

📄 VM instances

📄 Instance templates

👤 Sole-tenant nodes

📄 Machine images

🖨️ TPUs

📊 Committed use discou...

📅 Reservations

🔄 Migrate to Virtual Mach...

Storage

📄 Disks

🛒 Marketplace

📄 Release Notes

<|

Instances

Observability

Instance schedules

VM instances

Filter Enter property name or value

☐	Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐	🟢	lamp-1-vm	us-east1-c			10.142.0.2 (nic0)	104.196.99.84 (nic0)	SSH

Related actions

Copied

Show

🌤️ 9+ 16°C Clear

🪟 🔍 Search

🖱️ 📄 🗨️ 📁 🔄 🛒 🔒 📄

⬆️ ENG IN 🔊 🔌 📶 07:33 PM 27-11-2025 🔔



debian

Apache2 Debian Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Debian systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.



🏠 > Develop Your Google Cloud Network > Cloud Monitoring: Qwik Start

Contents

Cloud Monitoring: Qwik Start



Lab setup instructions and requirements >

End Lab 00:39:31

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked.
[Learn more.](#)

[Open Google Cloud console](#)

Username
student-02-44c3c12d53fd

Password

MEB8TZKXavPu

Project ID
qwiklabs-gcp-01-0bae0d0f

Create a Monitoring Metrics Scope

Set up a Monitoring Metrics Scope that's tied to your Google Cloud Project. The following steps create a new account that has a free trial of Monitoring.

- In the Cloud Console, click **Navigation menu** (☰) > View All Products > Observability > **Monitoring**.

When the Monitoring **Overview** page opens, your metrics scope project is ready.

Install the Monitoring and Logging agents

Agents collect data and then send or stream info to Cloud Monitoring in the Cloud Console.

The *Cloud Monitoring agent* is a collected-based daemon that gathers system and application metrics from virtual machine instances and sends them to Monitoring. By default, the Monitoring agent collects disk, CPU, network, and process metrics.

Lab instructions and tasks

- GSP089
 - Overview
 - Objectives
 - Setup and requirements
 - Task 1. Create a Compute Engine instance
 - Task 2. Add Apache2 HTTP Server to your instance
 - Task 3. Create an uptime check
 - Task 4. Create an alerting policy
 - Task 5. Create a

[← Previous](#)

Next >

Dashboard – qwiklabs-gcp-01-0ba...

Overview – Monitoring – qwiklab...

Apache2 Debian Default Page: It wor...

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring?referrer=search&project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅

🖨

👤 2

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰

Google Cloud

🔗 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦

🔧

🟢 3

?

⋮

🟡 S

📊 Observability Monitoring

📌

Overview

🎓 Learn

🏠 Overview

📊 Dashboards

📈 Application monitoring

Explore

📊 Metrics explorer

☰ Logs explorer

🔍 Log analytics

📊 Trace explorer

📊 Cost explorer

Detect

🔔 Alerting

⚙️ Error reporting

🕒 Uptime checks

🔗 Observability Scopes
qwiklabs-gcp-01-0bae0d009b16

📄 Release Notes

<|

Set up your project's observability

0% complete

Alerts
0
[View alerts](#)

Services with SLO out of budget
0
[View services](#)

Failed uptime checks
0
[View checks](#)

VMs with Ops Agent
1 of 1
No legacy agents. [View VMs](#)

Predefined

Last 1 hour

IST

Customize

Get Started with Cloud Observability

Google Cloud Observability offers a full suite of features for monitoring and troubleshooting your workloads, from logs, dashboards, alerts, and traces to SLO and uptime checks. You can customize this page to quickly access the views that matter most to you.

Logs

An at-a-glance view of the health of your applications. [View all logs here.](#)

Widget
Compute Engine Logs by Severity

Dashboards and Integrations

Google Cloud Observability provides [out-of-the-box dashboards](#) for a wide range of service types, including Google Cloud products like Compute Engine, Kubernetes Engine, Cloud Run, Cloud SQL, BigQuery, and Anthos along with third party integrations like PostgreSQL and ElasticSearch.

9+ 16°C Clear

🪟 🔍 Search

🖱️ 📁 🗂️ 📄 📧 📅 📌 📎 📏 📐 📊 📈 📉 📊 📈 📉

ENG IN

07:36 PM 27-11-2025

Q

Dashboard

Catalog

Paths

Collections

Develop Your Google Cloud Network > Cloud Monitoring: Qwik Start

Contents

Cloud Monitoring: Qwik Start

Lab setup instructions and requirements >

End Lab 00:36:52

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username student-02-44c3c12d53fdi

Password MEB8TZKXavPu

Project ID qwiklabs-gcp-01-0bae0d0f

sudo bash add-google-cloud-ops-agent-repo.sh --also-install

2. If asked if you want to continue, press Y.

3. Run the Logging agent install script command in the SSH terminal of your VM instance to install the Cloud Logging agent:

sudo systemctl status google-cloud-ops-agent "*"

Press q to exit the status.

sudo apt-get update

Task 3. Create an uptime check

Uptime checks verify that a resource is always accessible. For practice, create an uptime check to verify your VM is up.

75/100

Lab instructions and tasks

Overview

Objectives

Setup and requirements

Task 1. Create a Compute Engine instance

Task 2. Add Apache2 HTTP Server to your instance

Task 3. Create an uptime check

Task 4. Create an alerting policy

Task 5. Create a dashboard and chart

Task 6. View your logs

< Previous

Next >

1000 JOURNAL OF CLIMATE

```
Get:26 https://packages.cloud.google.com/apt cloud-sdk-bookworm/main amd64 Packages [4387 kB]
Get:27 https://packages.cloud.google.com/apt google-cloud-ops-agent-bookworm-2/main amd64 Packages [45.3 kB]
Fetched 32.6 MB in 4s (7766 kB/s)
```

```
Reading package lists... Done
```

```
student-02-44c3c12d53fd@lamp-1-vm:~$ sudo systemctl status google-cloud-ops-agent"
```

- `google-cloud-ops-agent.service` - Google Cloud Ops Agent

```
Loaded: loaded (/lib/systemd/system/google-cloud-ops-agent.service; enabled; preset: enabled)
```

Active: active (exited) since Thu 2025-11-27 13:59:31 UTC; 8min ago

```
Main PID: 1428 (code=exited, status=0/SUCCESS)
```

CPU: 195ms

```
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1404]: service:
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1404]: pipelines:
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1404]: default_pipeline:
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1404]: receivers: [hostmetrics]
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1404]: processors: [metrics filter]
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1404]: 2025/11/27 13:59:31 [Ports Check] Result: PASS
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1404]: 2025/11/27 13:59:31 [Network Check] Result: PASS
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1404]: 2025/11/27 13:59:31 [API Check] Result: PASS
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1404]: 2025/11/27 13:59:31 Startup checks finished
Nov 27 13:59:31 lamp-1-vm systemd[1]: Finished google-cloud-ops-agent.service - Google Cloud Ops Agent.
```

- `google-cloud-ops-agent-fluent-bit.service` - Google Cloud Ops Agent - Logging Agent

```
Loaded: loaded (/lib/systemd/system/google-cloud-ops-agent-fluent-bit.service; static)
```

```
Active: active (running) since Thu 2025-11-27 13:59:31 UTC; 8min ago
```

```
Main PID: 1478 (google cloud op)
```

Tasks: 31 (limit: 4682)

```
Memory: 33.4M
```

CPU: 1.790s

```
CGroup: /system.slice/google-cloud-ops-agent-fluent-bit.service
```

```
1478 /opt/google-cloud-ops-agent/libexec/google_cloud_ops_agent_wrapper -config_path /etc/google-cloud-ops-agent/config.yaml -log_path /var/log/google-cloud-ops-agent/ops-agent.log -log_level info -log_max_size 100M -log_max_age 7 -log_max_backups 3 -log_rotate_when size,age,backup
1509 /opt/google-cloud-ops-agent/subagents/fluent-bit/bin/fluent-bit --config /run/google-cloud-ops-agent-fluent-bit/fluent-bit.conf --parser /run/google-cloud-ops-agent-fluent-bit/fluent-bit.conf --log-path /var/log/google-cloud-ops-agent/ops-agent.log --log-level info --log-max-size 100M --log-max-age 7 --log-max-backups 3 --log-rotate-when size,age,backup
```

```
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]: processors:
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]:   metrics_filter:
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]:     type: exclude_metrics
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]:     metrics_pattern: []
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]: service:
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]:   pipelines:
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]:     default_pipeline:
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]:       receivers: [hostmetrics]
Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]:       processors: [metrics_filter]
Nov 27 13:59:31 lamp-1-vm systemd[1]: Started google-cloud-ops-agent-fluent-bit.service - Google Cloud Ops Agent - Logging Agent.
```

- `google-cloud-ops-agent-opentelemetry-collector.service` - Google Cloud Ops Agent - Metrics Agent

```
lines 1-40...skipping...
```

- `google-cloud-ops-agent.service` - Google Cloud Ops Agent

```
Loaded: loaded (/lib/systemd/system/google-cloud-ops-agent.service; enabled; preset: enabled)
```

```
Active: active (exited) since Thu 2025-11-27 13:59:31 UTC; 8min ago
```

```
Main PID: 1428 (code=exited, status=0/SUCCESS)
```

CPU: 195ms

ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-01-0bae0d009b16/zones/us-east1-c/instances/lamp-1-vm?authuser=0&hl=en_US&projectNumber=655743958698&useAdminProxy=true

SSH-in-browser

UPLOAD FILE

DOWNLOAD FILE

CPU: 1.790s

CGroup: /system.slice/google-cloud-ops-agent-fluent-bit.service

└─1478 /opt/google-cloud-ops-agent/libexec/google_cloud_ops_agent_wrapper -config_path /etc/google-cloud-ops-agent/config.yaml -log_path /var/log/google-cloud-ops-agent/subagents/1

ogging-module.log /o

└─1509 /opt/google-cloud-ops-agent/subagents/fluent-bit/bin/fluent-bit --config /run/google-cloud-ops-agent-fluent-bit/fluent_bit_main.conf --parser /run/google-cloud-ops-agent-flu

ent-bit/fluent_bit_p

Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]: processors:

Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]: metrics_filter:

Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]: type: exclude_metrics

Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]: metrics_pattern: []

Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]: service:

Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]: pipelines:

Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]: default_pipeline:

Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]: receivers: [hostmetrics]

Nov 27 13:59:31 lamp-1-vm google_cloud_ops_agent_engine[1430]: processors: [metrics_filter]

Nov 27 13:59:31 lamp-1-vm systemd[1]: Started google-cloud-ops-agent-fluent-bit.service - Google Cloud Ops Agent - Logging Agent.

● google-cloud-ops-agent-opentelemetry-collector.service - Google Cloud Ops Agent - Metrics Agent

Loaded: loaded (/lib/systemd/system/google-cloud-ops-agent-opentelemetry-collector.service; static)

Active: active (running) since Thu 2025-11-27 13:59:31 UTC; 8min ago

Main PID: 1476 (otelopscol)

Tasks: 8 (limit: 4682)

Memory: 45.2M

CPU: 1.414s

CGroup: /system.slice/google-cloud-ops-agent-opentelemetry-collector.service

└─1476 /opt/google-cloud-ops-agent/subagents/opentelemetry-collector/otelopscol --config=/run/google-cloud-ops-agent-opentelemetry-collector/otel.yaml --feature-gates=receiver.prom

etheusreceiver.Remove

Nov 27 13:59:32 lamp-1-vm otelopscol[1476]: 2025-11-27T13:59:32.291Z info otel@v0.136.0/parser_collection.go:646 one or more paths were modified to include their context pr

efix, please rewrite

Nov 27 13:59:32 lamp-1-vm otelopscol[1476]: 2025-11-27T13:59:32.295Z info service@v0.136.0/service.go:239 Starting otelopscol... {"resource": {"service.instance.id":

"1a405311-7923-47d6"

student-02-44c3c12d53fd@lamp-1-vm:~\$ sudo apt-get update

Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]

Get:5 file:/etc/apt/mirrors/debian-security.list Mirrorlist [39 B]

Hit:2 https://deb.debian.org/debian bookworm InRelease

Hit:3 https://deb.debian.org/debian bookworm-updates InRelease

Hit:7 https://packages.cloud.google.com/apt google-cloud-ops-agent-bookworm-all InRelease

Hit:4 https://deb.debian.org/debian bookworm-backports InRelease

Hit:6 https://deb.debian.org/debian-security bookworm-security InRelease

Hit:8 https://packages.cloud.google.com/apt google-compute-engine-bookworm-stable InRelease

Hit:9 https://packages.cloud.google.com/apt cloud-sdk-bookworm InRelease

Hit:10 https://packages.cloud.google.com/apt google-cloud-ops-agent-bookworm-2 InRelease

Reading package lists... Done

student-02-44c3c12d53fd@lamp-1-vm:~\$

9+ 16°C Clear

Search

ENG IN

07:39 PM 27-11-2025

Dashboard

Catalog

Paths

Collections

Cloud Monitoring: Qwik Start

Lab setup instructions and requirements

End Lab00:34:11

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked.
[Learn more.](#)

Open Google Cloud console

Username
student-02-44c3c12d53fdi

Password
MEB8TZKXavPu

Project ID
qwiklabs-gcp-01-0bae0d0f

Task 3. Create an uptime check

Uptime checks verify that a resource is always accessible. For practice, create an uptime check to verify your VM is up.

1. In the Cloud Console, in the left menu, click **Uptime checks**, and then click **Create Uptime Check**.

2. For **Protocol**, select HTTP.

3. For **Resource Type**, select **Instance**.

4. For **Instance**, select **lamp-1-vm**.

5. For **Check Frequency**, select **1 minute**.

6. Click **Continue**.

7. In Response Validation, accept the defaults and then click **Continue**.

8. In Alert & Notification, accept the defaults, and then click **Continue**.

75/100

Lab instructions and tasks

GSP089

Overview

Objectives

Setup and requirements

Task 1. Create a Compute Engine instance

Task 2. Add Apache2 HTTP Server to your instance

Task 3. Create an uptime check

Task 4. Create an alerting policy

Task 5. Create a dashboard and chart

Task 6. View your logs

< Previous

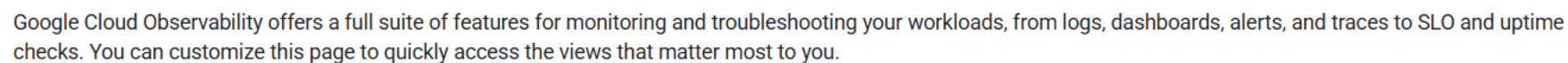
Next >

Set up your project's observability  0% complete

No legacy agents. [View VMs](#)

 **Customize**

x



An at-a-glance view of the health of your applications. [View all logs here.](#)

- Widget

Google Cloud Observability provides [out-of-the-box dashboards](#) for a wide range of service types, including Google Cloud products like Compute Engine, Kubernetes Engine, Cloud Run, Cloud SQL, BigQuery, and Anthos along with third party integrations like PostgreSQL and ElasticSearch.

Dashboard – qwiklabs-gcp-01-0ba...

Monitoring – qwiklabs-gcp-01-0...

Apache2 Debian Default Page: It wor...

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/uptime/create?project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅

🖨

👤 2

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰

Google Cloud

🔍 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦

🔧

3

?

⋮

S

📊 Observability Monitoring

📊 Cost explorer

Detect

🔔 Alerting

📄 Error reporting

🖥 Uptime checks

🔗 Synthetic monitoring

📈 SLOs

Configure

🔗 Integrations

📄 Log-based metrics

🔗 Log router

📄 Logs storage

📈 Metrics management

🔗 Observability Scopes

📄 Release Notes

<I

← Create Uptime Check

• Target

• Response Validation (optional)

• Alert & Notification (optional)

• Review

Create

Cancel

Select the resource to be monitored.

Protocol

HTTP

Resource Type

Instance

Applies to

☒ Single

Apply to one resource.

☐ Group

Apply to a collection of resources that is defined as a Monitoring Group.

Instance*

lamp-1-vm

Path

Check Frequency

1 minute

More target options

Continue

🌤 16°C

Clear

🪟

🔍 Search

🖱

📁

💬

📁

🌐

🛒

🔒

🔊

🔌

📶

ENG IN

07:42 PM

27-11-2025

🔔

Dashboard – qwiklabs-gcp-01-0ba...

Monitoring – qwiklabs-gcp-01-0...

Apache2 Debian Default Page: It wor...

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/uptime/create?project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅

🖨

👤 2

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰

Google Cloud

🔗 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦

🔧

3

?

⋮

S

📊 Observability Monitoring

📊 Cost explorer

Detect

🔔 Alerting

📄 Error reporting

🖥 Uptime checks

🔗 Synthetic monitoring

📈 SLOs

Configure

🔗 Integrations

📄 Log-based metrics

🔗 Log router

📄 Logs storage

📈 Metrics management

🔗 Observability Scopes

📄 Release Notes

<|

🔙

Create Uptime Check

✔ Target

✔ Response Validation (optional)

✔ Alert & Notification (optional)

• Review

Create

Cancel

Enter a name for the uptime check.

Title *

Lamp Uptime Check

User Labels ?

Key

Value (optional)

+ Add user label

^ Hide user labels

Test uptime check (optional)

We recommend that you test your uptime check before creating.

✔ Responded with "200 (OK)" in 240 ms.

Test

View Details

9+ 16°C Clear

🪟 🔍 Search 🖱 📁 🗂 🌐 🛒 🔒 📄

⬆️ 📶 🔊 🔌

ENG IN 07:42 PM 27-11-2025 🔔

Dashboard - qwiklabs-gcp-01-0ba...

Uptime checks - Monitoring - q...

Apache2 Debian Default Page: It wor...

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/uptime?project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅

📄

🗨️ 2

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰

Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦

📄

🟢 3

?

⋮

👤 S

📊 Observability Monitoring

📊 Cost explorer

🔍 Detect

🔔 Alerting

📄 Error reporting

📊 Uptime checks

👤 Synthetic monitoring

📄 SLOs

⚙️ Configure

🔗 Integrations

📄 Log-based metrics

🔗 Log router

📄 Logs storage

📊 Metrics management

📄 Observability Scopes

📄 Release Notes

<|

📌 Uptime checks

📄 + Create uptime check

+ 1 recommended alert

📖 Learn

🌟

Try the new Synthetic monitoring capability! New

Proactively monitor your application's critical user journeys by simulating user behavior. Detect and alert on failures, and receive detailed reports for debugging. [Learn more.](#)

Create Synthetic Monitor

View Documentation

Filter

Enter property name or value

Display Name ↑

Asia Pacific

Europe

North America

South America

Policies

Lamp Uptime Check

1

🌤️ 16°C Clear

🔍 Search

🖱️

🗂️

💬

📁

🌀

🛒

🔥

📄

⬆️

ENG IN

📶

🔊

🔋

07:43 PM 27-11-2025

🔔

Task 4. Create an alerting policy

Use Cloud Monitoring to create one or more alerting policies.

1. In the left menu, click **Alerting**, and then click **+Create Policy**.
2. Click on **Select a metric** dropdown. Uncheck the **Active**.
3. Type **Network traffic** in filter by resource and metric name and click on **VM instance > Interface**. Select **Network traffic** (agent.googleapis.com/interface/traffic) and click **Apply**. Leave all other fields at the default value.
4. Click **Next**.
5. Set the **Threshold position** to **Above threshold**, **Threshold value** to **500** and **Advanced Options > Retest window** to **1 min**. Click **Next**.
6. Click on the drop down arrow next to **Notification Channels**, then click on **Manage Notification Channels**.

A **Notification channels** page will open in a new tab.

Dashboard - qwiklabs-gcp-01-0ba...

Uptime checks - Monitoring - q...

Apache2 Debian Default Page: It wor...

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/uptime?project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅

🖨

👤 2

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰

Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦

🔧

🟢 3

?

⋮

👤 S

📊 Observability Monitoring

📊 Cost explorer

🔍 Detect

🔔 Alerting

🔧 Error reporting

📺 Uptime checks

👤 Synthetic monitoring

📊 SLOs

⚙️ Configure

🔗 Integrations

📄 Log-based metrics

🔗 Log router

📄 Logs storage

📈 Metrics management

📊 Observability Scopes

📄 Release Notes

📌 Uptime checks

📄 + Create uptime check

+ 1 recommended alert

📖 Learn

🌟

Try the new Synthetic monitoring capability! New

Proactively monitor your application's critical user journeys by simulating user behavior. Detect and alert on failures, and receive detailed reports for debugging. [Learn more.](#)

Create Synthetic Monitor

View Documentation

🔍 Filter Enter property name or value

Display Name ↑	Asia Pacific	Europe	North America	South America	Policies	
Lamp Uptime Check	●	●	●	●	1	⋮

🔗 https://console.cloud.google.com/monitoring/alerting?project=qwiklabs-gcp-01-0bae0d009...

🌤 16°C Clear

🔍 Search

🖨

🗂

💬

📁

🌀

🛒

🔥

📄

⬆

ENG IN

📶

🔊

🔌

🕒 07:43 PM 27-11-2025

🔔

Dashboard – qwiklabs-gcp-01-0ba...

Monitoring – qwiklabs-gcp-01-0...

Apache2 Debian Default Page: It wor...

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/alerting/policies/create?project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅

📄

👤 2

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

🔗 qwiklabs-gcp-01-0bae0d009b16

🔍 Search (/) for resources, docs, products, and more

🔍 Search

✦

📄

🟢 3

?

⋮

🟡 S

← Create alerting policy

+ Add alert condition

🗑 Delete alert condition

🔗 View Code

ALERT CONDITIONS

New condition

Configure trigger

ALERT DETAILS

Notifications and name

Review alert

Policy configuration mode

☒ Builder

☐ Code editor (MQL or PromQL)

Select a metric

🔍 Network traffic

Active

POPULAR RESOURCES

Amazon EC2 Instance3 metrics >

VM Instance6 metrics >

ACTIVE RESOURCES

GCP Location1 metric >

VM Instance6 metrics >

INACTIVE RESOURCES

Amazon EC2 Instance3 metrics >

ACTIVE METRIC CATEGORIES

Interface1 metric >

INACTIVE METRIC CATEGORIES

Popular metrics

Elasticsearch1 metric >

Instance3 metrics >

Memcached1 metric >

ACTIVE METRICS

Network traffic
agent.googleapis.com/interface/traffic

INACTIVE METRICS

Selection preview

VM Instance > interface

Cancel

Apply

Looking to alert on other types of data?

Create a log-based alert policy

Create a SQL alert policy

by querying and aggregating log data
Observability Analytics.

SQL alert

Learn more

9+ 16°C Clear

🪟 🔍 Search

🖱️ 📁 📄 📅 📧 📧 📧 📧 📧

⬆️ ENG IN 📶 🔊 🔋 07:44 PM 27-11-2025 🔔

Dashboard – qwiklabs-gcp-01-0ba...

Monitoring – qwiklabs-gcp-01-0...

Apache2 Debian Default Page: It wor...

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/alerting/policies/create?project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅 📄 🌙 2 ☰

🔖 Free AI Paraphrasin...

📄 Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

🔗 qwiklabs-gcp-01-0bae0d009b16

🔍 Search (/) for resources, docs, products, and more

🔍 Search

✦ 📄 3 ? ⋮

🔴 S

← Create alerting policy

+ Add alert condition

🗑 Delete alert condition

🔗 View Code

ALERT CONDITIONS

VM Instance - Network traffic

⬆

• Configure trigger

ALERT DETAILS

• Notifications and name

• Review alert

Policy configuration mode

☒ Builder

☐ Code editor (MQL or PromQL)

Select a metric ⓘ

📊 VM Instance - Network traffic ▾

Add filters Optional

Selections made on the chart do not affect the alert policy [Learn more](#) 🔗

Add a filter

Transform data

Within each time series ⓘ

Rolling window *

5 min

⌵

Adjust the length of time a signal is calculated for. Example: Mean of CPU utilization for 5 minutes is above 80%

✓ 1 hour

6 hours

1 day

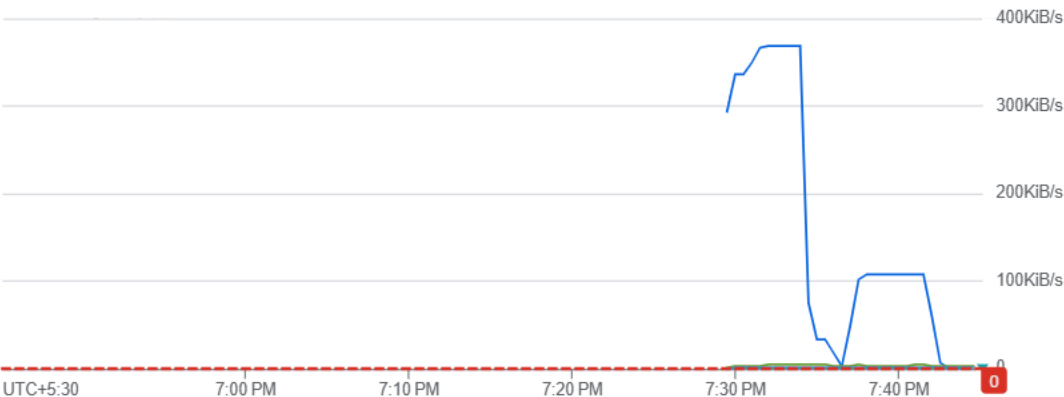
7 days

30 days

Custom ▾

VM Instance - Network traffic ⓘ

📊 ⋮



Filter

Enter property name or value

ⓘ

☰

✕

☐	device	↑	direction	Value
☐	ens4	•	rx	329.371 B/s
☐	ens4	■	tx	2.599 KiB/s
☐	lo	◆	rx	405.144 B/s

Create Policy

Provide feedback

Cancel

9+ 16°C Clear

🪟 🔍 Search

🖱 📁 🗂 📄 📅 📧 📧

⬆ ENG IN 📶 🔊 🔌

07:45 PM 27-11-2025 🔔

Dashboard – qwiklabs-gcp-01-0ba...

Monitoring – qwiklabs-gcp-01-0...

Apache2 Debian Default Page: It worl

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/alerting/policies/create?project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅 📄 🌙 2 ☰

🔗 Free AI Paraphrasin...

📄 Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

🔗 qwiklabs-gcp-01-0bae0d009b16

🔍 Search (/) for resources, docs, products, and more

🔍 Search

✦ 📄 3 ? ⋮

👤 S

⬅️ Create alerting policy

+ Add alert condition

🗑️ Delete alert condition

🔗 View Code

ALERT CONDITIONS

✓ VM Instance - Network traffic

⬆️

• Configure trigger

ALERT DETAILS

• Notifications and name

• Review alert

Alert trigger

Any time series violates

▼

Threshold position

Above threshold

▼

Threshold value

500

B/s

Advanced Options

⬆️

Retest window *

1 min

▼

Used to retest alerts. Expand timeframe if alerts have a tendency to be false positive.

Evaluation of missing data

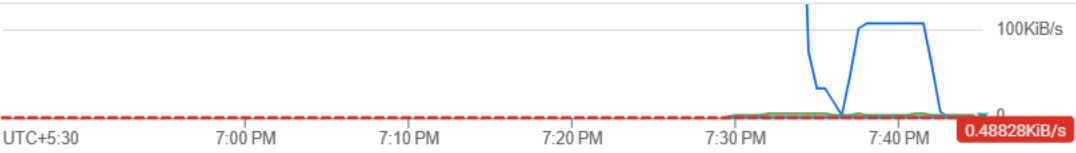
▼ ?

Choose how conditions are evaluated when data is missing.

Condition name *

VM Instance - Network traffic

Next



Filter

Enter property name or value

?

⌵

☐	device	↑	direction	Value
☐	ens4	•	rx	398.101 B/s
☐	ens4	■	tx	3.217 KiB/s
☐	lo	◆	rx	499.568 B/s

Rows per page:

3 ▼

1 – 3 of 4

⬅️ ➡️

Estimated monthly cost

Coming 2026

~~\$0.22~~ \$0.00

Pricing will be introduced in 2026. Estimates do not account for discounts, credits, and the free tier. [Learn more](#)

✓ Show cost breakdown

Create Policy

Provide feedback

Cancel

🌤️ 9+ 16°C Clear

🪟 🔍 Search

🖱️ 🗂️ 🗨️ 📁 🌐 🛒 🔒 📅

⬆️ ENG IN 📶 🔊 🔌 07:45 PM 27-11-2025 🔔

Dashboard

Catalog

Paths

Collections

Cloud Monitoring: Qwik Start

Contents

Lab setup instructions and requirements

End Lab00:27:11

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-02-44c3c12d53fdi

Password

MEB8TZKXavPu

Project ID

qwiklabs-gcp-01-0bae0d0f

Manage Notification Channels.

A Notification channels page will open in a new tab.

7. Scroll down the page and click on **ADD NEW** for **Email**.

8. In the **Create Email Channel** dialog box, enter your personal email address in the **Email Address** field and a **Display name**.

9. Click on **Save**.

10. Go back to the previous **Create alerting policy** tab.

11. Click on **Notification Channels** again, then click on the **Refresh icon** to get the display name you mentioned in the previous step.

12. Click on **Notification Channels** again if necessary, select your **Display name** and click **OK**.

13. Add a message in documentation, which will be included in the emailed alert.

14. Mention the **Alert name** as `Inbound Traffic Alert`.

15. Click **Next**.

75/100

Lab instructions and tasks

GSP089

Overview

Objectives

Setup and requirements

Task 1. Create a Compute Engine instance

Task 2. Add Apache2 HTTP Server to your instance

Task 3. Create an uptime check

Task 4. Create an alerting policy

Task 5. Create a dashboard and chart

Task 6. View your logs

< Previous

Next >

Dashboard – qwiklabs-gcp-01-0ba...

Notification channels – Monitori

Apache2 Debian Default Page: It wor

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/alerting/policies/create?project=qwiklabs-gcp-01-0bae0d009b16

🔍

📄

🗨️ 2

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

qw

← Create alerting policy

ALERT CONDITIONS

✓ VM Instance - Network traffic

✓ Configure trigger

ALERT DETAILS

• Notifications and name

• Review alert

Notification channels

Google Chat Preview

No Google Chat channels configured

Add New

PagerDuty Services

No PagerDuty services configured

Add New

PagerDuty Sync Beta

No PagerDuty Sync channels configured

Add New

Slack ?

No Slack channels configured

Add New

Webhooks

No webhook channels configured

Add New

Email

No emails configured

Add New

SMS

No SMS channels configured

Add New

Pub/Sub

No Pub/Sub channels configured

Add New

Create Email Channel

Email addresses can be set to receive notifications from your alerting when a new incident is created.

Email Address *

devt40363@gmail.com

Display Name *

Lamp Notification

Cancel

Save

9+ 16°C Clear

🪟 🔍 Search 🖱️ 📁 🗨️ 🌐 🛒 🔒 📅

⬆️ ENG IN 📶 🔊 🔋 07:47 PM 27-11-2025 🔔

Dashboard – qwiklabs-gcp-01-0ba...

Monitoring – qwiklabs-gcp-01-0...

Apache2 Debian Default Page: It worl

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/alerting/policies/create?project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅

📄

👤 2

☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦

📄

🟢 3

?

⋮

🟡 S

← Create alerting policy

+ Add alert condition

🗑 Delete alert condition

🔗 View Code

ALERT CONDITIONS

✓ VM Instance - Network traffic

✓ Configure trigger

ALERT DETAILS

• Notifications and name

• Review alert

Configure notifications and finalize alert

Configure notifications Recommended

Use notification channel

Notification Channels

Filter Type to filter

Email

✓ Lamp Notification

Manage Notification Channels

Cancel

OK

redundant channel.

Learn more

Notify on incident closure

Incident autoclose duration

Create Policy

Provide feedback

Cancel

9+ 16°C Clear

🪟 🔍 Search 🖱 📁 📄 🌐 🛒 🔒 📅

ENG IN 📶 🔊 🔌 07:47 PM 27-11-2025 🔔

Dashboard – qwiklabs-gcp-01-0ba...

Monitoring – qwiklabs-gcp-01-0...

Apache2 Debian Default Page: It worl

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/alerting/policies/create?project=qwiklabs-gcp-01-0bae0d009b16

🔍

📅

📄

👤 2

☰

🔗 Free AI Paraphrasin...

📄 Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

🔗 qwiklabs-gcp-01-0bae0d009b16

🔍 Search (/) for resources, docs, products, and more

🔍 Search

✦

📄

3

?

⋮

S

← Create alerting policy

+ Add alert condition

🗑 Delete alert condition

🔗 View Code

ALERT CONDITIONS

✓ VM Instance - Network traffic

✓ Configure trigger

ALERT DETAILS

• Notifications and name

• Review alert

Documentation Optional

Enter any documentation you would like included with the notification. You can use markdown, [variables](#), and channel-specific controls. Markdown formatting may not apply to all notification channels.

Text Field

Lamp Notification: Above Threshold

?

Name the alert policy

Alert policy name *

Inbound Traffic Alert

Next

Create Policy

Provide feedback

Cancel

9+ 16°C Clear

🪟 🔍 Search

🖱️ 📄 🗨️ 📁 🌐 🛒 🔒 📅

⬆️ ENG IN 📶 🔊 🔋

07:48 PM 27-11-2025 🔔

ALERT CONDITIONS

✓ VM Instance - Network traffic

✓ Configure trigger

ALERT DETAILS

✓ Notifications and name

• Review alert

VM Instance - Network traffic

Condition type	Triggers when	Threshold position	Threshold value	Retest window
Threshold	Any time series cross threshold	Above threshold	500	1 min

📘

📈

⋮

UTC+5:30 6:55 PM 7:00 PM 7:05 PM 7:10 PM 7:15 PM 7:20 PM 7:25 PM 7:30 PM 7:35 PM 7:40 PM 7:45 PM

0.48828KiB/s

🔍 Filter

Enter property name or value

🔍

⋮

✕

☐	device	↑	direction	Value
☐	ens4	●	rx	298.65 B/s
☐	ens4	■	tx	2.315 KIB/s
☐	lo	◆	rx	355.309 B/s

Rows per page: 3 1 – 3 of 4 < >

Policy Details

Alert Policy Name

Inbound Traffic Alert

Severity

No severity

Notification channels

Notify on incident open only.

Incidents with absent data will auto close after 1 minute.

Display name	Type
Lamp Notification	Email

Documentation

Lamp Notification: Above Threshold

Dashboard – qwiklabs-gcp-01-0ba...

Policy details – Monitoring – qwi...

Apache2 Debian Default Page: It wor...

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/alerting/policies/16494762412099309108?project=qwiklabs-gcp-01-0bae0d009b16

🔍 | 📱

📅 | 🖨️ | 🗨️ 2 | ☰

📄 Free AI Paraphrasin...

📄 Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

🔍 Search (/) for resources, docs, products, and more

🔍 Search

🌟 📺 3 ? ⋮ 📌

📺 Observability Monitoring

📄 Overview

📊 Dashboards

📈 Application monitoring

Explore

📊 Metrics explorer

📄 Logs explorer

🔍 Log analytics

📄 Trace explorer

📊 Cost explorer

Detect

🔔 Alerting

🔧 Error reporting

📺 Uptime checks

👤 Synthetic monitoring

📊 SLOs

👤 Observability Scopes

👤 qwiklabs-gcp-01-0bae0d009b16

📄 Release Notes

<|

← Policy details

✅ Enabled

✎ Edit

📄 Copy

🗑️ Delete

📄 JSON

✓ 1 hour 6 hours 1 day 1 week 1 month 6 weeks

Inbound Traffic Alert

Conditions Severity

Policy violates when ANY condition is met No severity

VM Instance - Network traffic

Condition type

Threshold

Triggers when

Any time series cross threshold

Threshold position

Above threshold

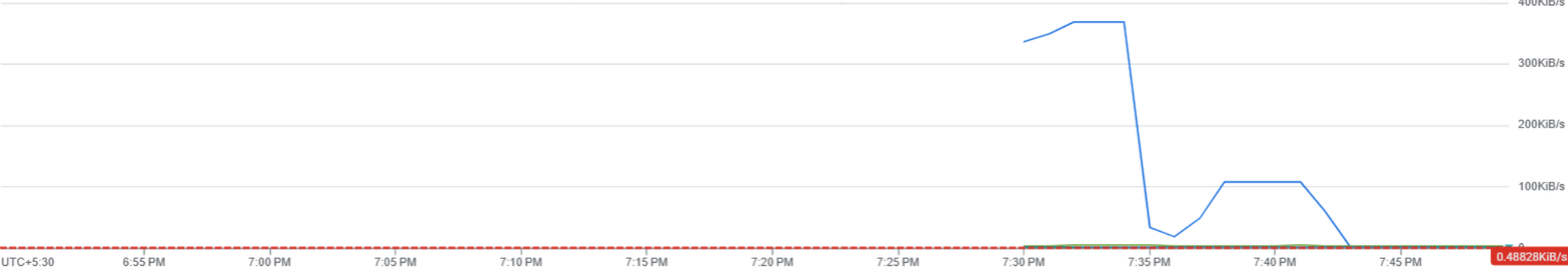
Threshold value

500 B/s

Retest window

1 min

📊



Filter

Enter property name or value

🔍

📄

✕

☐	device ↑	direction	Value
☐	ens4	rx	327.426 B/s
☐	ens4	tx	2.521 KiB/s

9+ 16°C Clear

🔍 Search

🖱️ 📄 🗨️ 📁 📊 📄 📄 📄

⬆️ ENG IN 📶 🔊 🔌 07:49 PM 27-11-2025 📌

Dashboard

Catalog

Paths

Collections

Cloud Monitoring: Qwik Start

Contents

Cloud Monitoring: Qwik Start

Lab setup instructions and requirements

End Lab00:23:53

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Usernamestudent-02-44c3c12d53fdi

PasswordMEB8TZKXavPu

Project IDqwiklabs-gcp-01-0bae0d0f

Cloud Monitoring: Qwik Start

Create an uptime check and alerting policy

Check my progress

Assessment Completed!

Task 5. Create a dashboard and chart

You can display the metrics collected by Cloud Monitoring in your own charts and dashboards. In this section you create the charts for the lab metrics and a custom dashboard.

- In the left menu select **Dashboards**, and then **+Create Custom Dashboard**.
- Name the dashboard `Cloud Monitoring LAMP Qwik Start Dashboard`.

Add the first chart

- Click on **+ ADD WIDGET**

75/100

Lab instructions and tasks

Task 5. Create a dashboard and chart

Dashboard – qwiklabs-gcp-01-0ba...

Policy details – qwi

Apache2 Debian Default Page: It wor

+

< > ↺ 🏠

🔖

🔗

https://console.cloud.google.com/monitoring/alerting/policies/16494762412099309108?project=qwiklabs-gcp-01-0bae0d009b16

🔍 | 📱

📅 | 🖨️ | 🌙 2 | ☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦ 📺 3 ? ⋮ S

Observability Monitoring 📌

🏠 Overview

📊 Dashboards

📈 Application monitoring

Explore ^

📊 Metrics explorer

📋 Logs explorer

🔍 Log analytics

📄 Trace explorer

💰 Cost explorer

Detect ^

🔔 Alerting

🔧 Error reporting

🕒 Uptime checks

👤 Synthetic monitoring

📊 SLOs

🔗 Observability Scopes > qwiklabs-gcp-01-0bae0d009b16

📄 Release Notes

← Policy details

✅ Enabled

✎ Edit

📄 Copy

🗑️ Delete

📄 JSON

✓ 1 hour 6 hours 1 day 1 week 1 month 6 weeks

Inbound Traffic Alert

Conditions Severity

Policy violates when ANY condition is met No severity

VM Instance - Network traffic

Condition type

Threshold

Triggers when

Any time series cross threshold

Threshold position

Above threshold

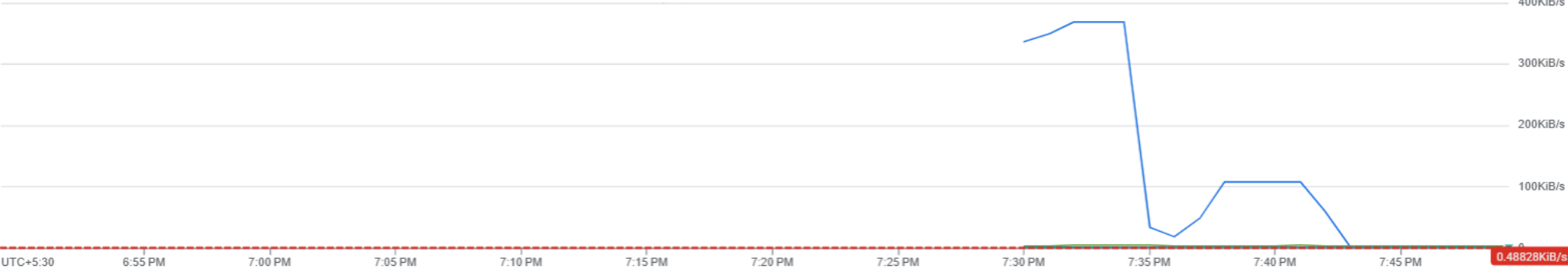
Threshold value

500 B/s

Retest window

1 min

📄



UTC+5:30

6:55 PM

7:00 PM

7:05 PM

7:10 PM

7:15 PM

7:20 PM

7:25 PM

7:30 PM

7:35 PM

7:40 PM

7:45 PM

0.48828 KiB/s

Filter

Enter property name or value

🔍

📄

✕

☐	device ↑	direction	Value
☐	ens4	rx	327.426 B/s
☐		tx	2.521 KiB/s

9+ 16°C Clear

🪟 🔍 Search 🖱️ 📁 📄 🗨️ 🌐 🛒 🔒 📅

⬆️ ENG IN 📶 🔊 🔋 07:52 PM 27-11-2025 🔔

Dashboard – qwiklabs-gcp-01-0ba...

Monitoring – qwiklabs-gcp-01-0...

Apache2 Debian Default Page: It wor...

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/dashboards?project=qwiklabs-gcp-01-0bae0d009b16

🔍 | 🛡️

📅 | 📄 | 🌙 2 | ☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

Search (/) for resources, docs, products, and more

🔍 Search

✦ 📄 3 ? ⋮ 📄

📊 Observability Monitoring

🏠 Overview

📊 Dashboards

📈 Application monitoring

Explore

📊 Metrics explorer

📋 Logs explorer

🔍 Log analytics

🔍 Trace explorer

📊 Cost explorer

Detect

🔔 Alerting

🔔 Error reporting

📺 Uptime checks

🔗 Synthetic monitoring

🔗 SLOs

🔗 Observability Scopes
qwiklabs-gcp-01-0bae0d009b16

📄 Release Notes

<|

Dashboards

Add or create more dashboards



Create Custom Dashboard

Create a custom dashboard to fit your needs.

Create Custom Dashboard



Dashboard Templates

Add a pre-made dashboard from the template catalog.

View Dashboard Templates



Import Dashboards

Upload your Grafana dashboards into Google Cloud.

Import Grafana Dashboards

My Dashboards

+ Labels

🗑 Delete

🏷 Tags

⚙ Manage Labels

🔍 Filter

Search your dashboards

🔍 ?

☰

☰ All	22	☐ ↓	Name	Type ?	Description ?	Actions
🕒 Recently Viewed	0	☐ ☆	Autoscaler Monitoring	Google Services	This dashboard has 2 charts for the related autoscaler metrics: Current...	🗑 ✓
★ Favorites	0	☐ ☆	Cluster Director Block Network	Google Services	This dashboard features 14 charts providing an in-depth view of Cluster...	🗑 ✓
👤 Recommended	22	☐ ☆	Cluster Director Health Monitoring	Google Services	This dashboard features 18 charts presenting detailed insights into...	🗑 ✓
▼ Integrations		☐ ☆	Cluster Director Transmission Efficiency	Google Services	This dashboard features 24 charts providing insights into Cluster Director...	🗑 ✓
Ops Agent	0	☐ ☆	Cluster Health Scanner	Google Services	This dashboard features charts for monitoring the results of the Cluster...	🗑 ✓
GMP	0	☐ ☆	Compute Engine - Instances	In-Context	This dashboard shows charts displaying metrics of VMs instances in a...	✓
▼ Type		☐ ☆	Disks	Google Services	This dashboard has charts displaying an inventory of GCE Disks, Volume...	🗑 ✓
Custom	0	☐ ☆	Firewalls	Google Services	This dashboard shows the list of firewalls in a project.	
Playbook	7	☐ ☆	GCE Interactive Playbook - Cluster Director Block Netwo...	Playbook	The playbook helps you monitor and troubleshoot network locality issues i...	🗑 ✓
Google Services	15	☐ ☆	GCE Interactive Playbook - Cluster Director Health Moni...	Playbook	This playbook helps you monitor, identify, and troubleshoot infrastructure...	🗑 ✓
		☐ ☆	GCE Interactive Playbook - Host Events	Playbook	This playbook will assist you with troubleshooting VM host events such as...	🗑 ✓
		☐ ☆	GCE Interactive Playbook - MIG Autoscaling Failures	Playbook	This olavbook will assist you with troubleshooting autoscaling issues in...	🗑 ✓

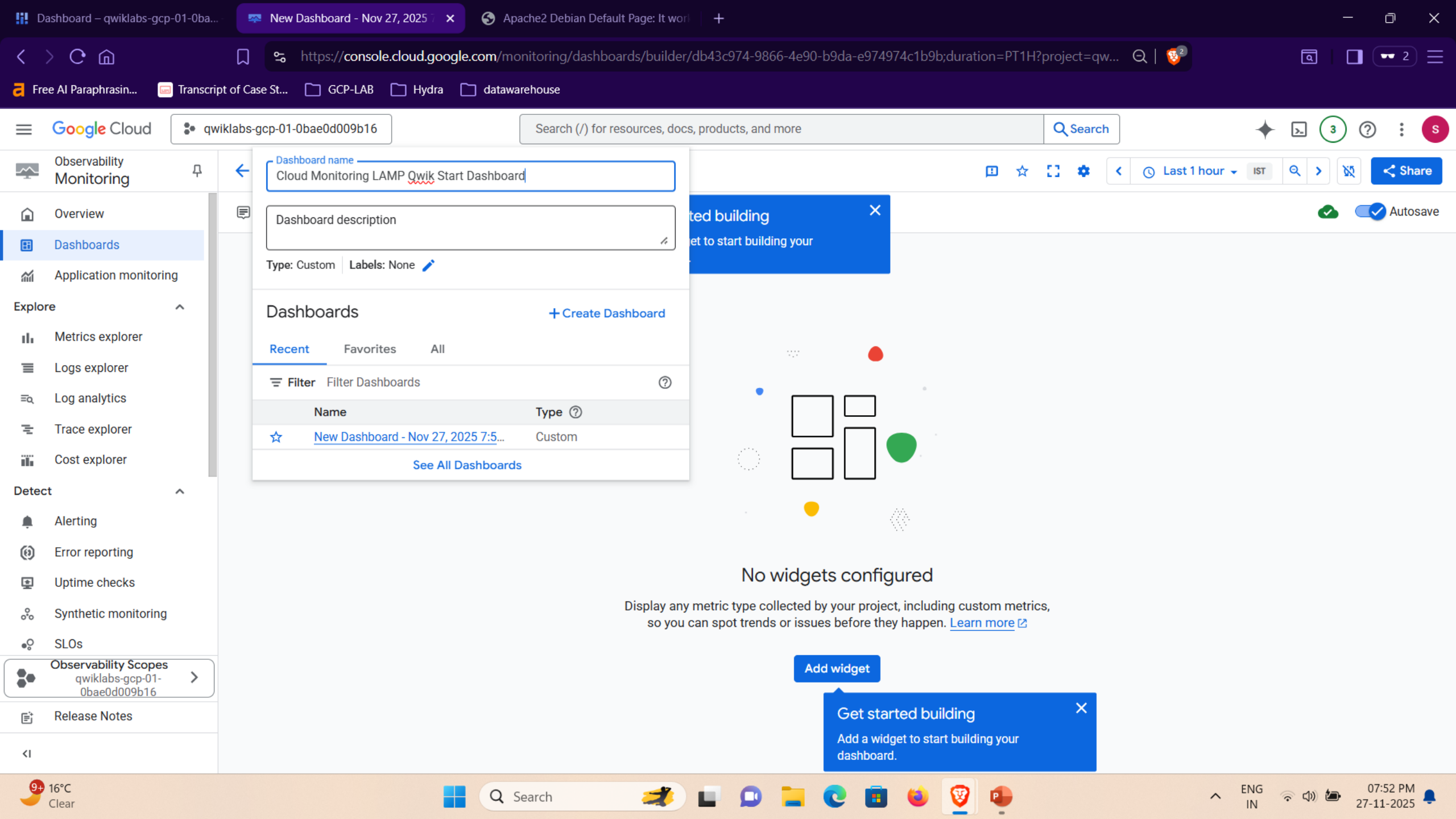
Rows per page: 50 1 – 22 of 22 < >

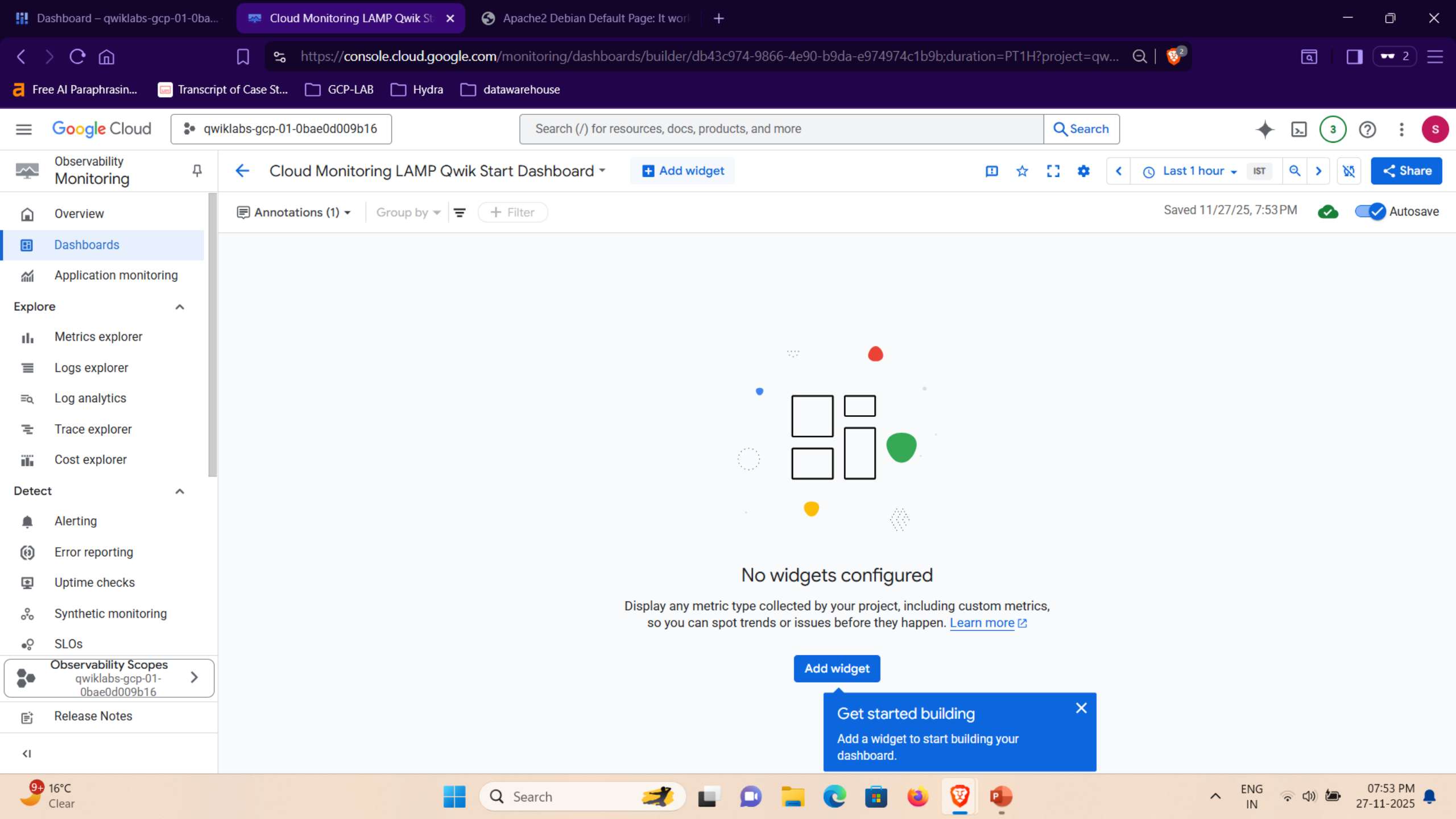
9+ 16°C Clear

🪟 🔍 Search

🖱 📁 📄 📅 📊 📈 📉 📋 📌 📍 📎 📏 📐 📑 📒 📓 📔 📕 📖 📗 📙 📚

ENG IN 📶 🔊 🔌 07:52 PM 27-11-2025





Dashboard – qwiklabs-gcp-01-0ba...

Cloud Monitoring LAMP Qwik St

Apache2 Debian Default Page: It worl

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/dashboards/builder/db43c974-9866-4e90-b9da-e974974c1b9b;duration=PT1H?project=qw...

🔍 | 🛡️

📅 | 📄 | 🗨️ 2 | ☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰

Configure widget

📄

🔗

⬇️

🔗

<

🕒 Last 1 hour

IST

🔍

>

🗑️

Apply

Cancel

Queries

+ Add query

➔ Create ratio

🔛 Auto-Run

A

Metric

Select a metric

?

Filter

Add filter

Aggregation

Unaggregated

by

None

+

<> PromQL

📄

🗑️

Results

📄

Select a metric

Reset

×

🔍 CPU load (1m)

Active

POPULAR RESOURCES

VM Instance2 metrics

ACTIVE RESOURCES

VM Instance2 metrics

INACTIVE RESOURCES

Amazon EC2 Instance1 metric

Bare Metal Solution Instance1 metric

ACTIVE METRIC CATEGORIES

Cpu1 metric

INACTIVE METRIC CATEGORIES

Popular metrics5 metrics

Guest1 metric

ACTIVE METRICS

CPU load (1m)
agent.googleapis.com/cpu/load_1m

INACTIVE METRICS

Selection preview

VM Instance > cpu

Cancel

Apply

Display

Widget type

Line chart

Widget title

Analysis Mode

Compare to past

Threshold Line

Y-axis scale

Widget Visibility

9+ 16°C Clear

🪟 🔍 Search 🐼 📁 🗨️ 🌐 🛒 🔒 📅

ENG IN

📶 🔊 🔌

07:53 PM 27-11-2025

🔔

Dashboard - qwiklabs-gcp-01-0ba...

Cloud Monitoring LAMP Qwik St

Apache2 Debian Default Page: It worl

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/dashboards/builder/db43c974-9866-4e90-b9da-e974974c1b9b;duration=PT1H?project=qw...

🔍 | 🛡️

📅 | 🖨️ | 🌙 2 | ☰

📄 Free AI Paraphrasin...

📄 Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

🔍 Search (/) for resources, docs, products, and more

🔍 Search

✦ 📄 3 ? ⋮

📊 Observability Monitoring

🏠 Overview

📊 Dashboards

📈 Application monitoring

Explore

📊 Metrics explorer

📄 Logs explorer

🔍 Log analytics

📄 Trace explorer

📊 Cost explorer

Detect

🔔 Alerting

⚙️ Error reporting

📄 Uptime checks

👤 Synthetic monitoring

📊 SLOs

👤 Observability Scopes

👤 qwiklabs-gcp-01-0bae0d009b16

📄 Release Notes

<I

🔙 Cloud Monitoring LAMP Qwik Start Dashboard

+ Add widget

📄 Annotations (2)

Group by

+ Filter

🕒 Last 1 hour

IST

🔍

🔔

🔗 Share

Saved 11/27/25, 7:54 PM

🟢 Autosave

CPU load (1m) [SUM]



Time	CPU load (1m) [SUM]
7:30 PM	0.55
7:31 PM	0.25
7:32 PM	0.35
7:33 PM	0.25
7:34 PM	0.15
7:35 PM	0.10
7:36 PM	0.20
7:37 PM	0.15
7:38 PM	0.10
7:39 PM	0.10
7:40 PM	0.15
7:41 PM	0.10
7:42 PM	0.10
7:43 PM	0.10
7:44 PM	0.10
7:45 PM	0.10
7:46 PM	0.10
7:47 PM	0.10
7:48 PM	0.10
7:49 PM	0.10
7:50 PM	0.10

UTC+5:30 7:00 PM 7:10 PM 7:20 PM 7:30 PM 7:40 PM 7:50 PM

● load_1m

🌤️ 16°C Clear

🔍 Search

🖨️ 📄 🗂️ 📁 📌 📄 📄 📄

ENG IN 📶 🔊 🔌

07:54 PM 27-11-2025 🛎️

 Last 1 hour ▾

 Apply

☒ Auto-Run [↗](#)

< > PromQL

Display

Widget Visibility

5 PM
2025

☰

Configure widget

Queries

A

Metric

Select a metric

?

Filter

Add filter

Aggregation

Unaggregated

by

None

+

<

🕒 Last 1 hour

IST

🔍

>

🗑️

Apply

Cancel

Auto-Run

↗

Results

Select a metric

Reset

×

Received Packets

Active

POPULAR RESOURCES

VM Instance4 metrics >

ACTIVE RESOURCES

VM Instance4 metrics >

INACTIVE RESOURCES

Amazon EC2 Instance1 metric >

Cloud NAT Gateway2 metrics >

Cloud Router1 metric >

Selection preview

VM Instance > instance > Received packets

ACTIVE METRIC CATEGORIES

Instance1 metric >

INACTIVE METRIC CATEGORIES

Popular metrics5 metrics >

Nat2 metrics >

Zookeeper1 metric >

ACTIVE METRICS

Received packets
compute.googleapis.com/instance/network/rece...

INACTIVE METRICS

Cancel

Apply

Display

Widget type

Line chart

?

Widget title

Received Packets

Analysis Mode

?

▼

Compare to past

?

▼

Threshold Line

?

▼

Y-axis scale

?

▼

Widget Visibility

?

▼

Dashboard – qwiklabs-gcp-01-0ba...

Cloud Monitoring LAMP Qwik St

Apache2 Debian Default Page: It worl

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/dashboards/builder/db43c974-9866-4e90-b9da-e974974c1b9b;duration=PT1H?project=qw...

🔍 | 🛡️

📅 | 🖨️ | 🌙 2 | ☰

📄 Free AI Paraphrasin...

📄 Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

🔍 Search (/) for resources, docs, products, and more

🔍 Search

✦ 📄 3 ? ⋮

📄

📊 Observability Monitoring

🏠 Overview

📊 Dashboards

📈 Application monitoring

Explore

📊 Metrics explorer

📋 Logs explorer

🔍 Log analytics

📊 Trace explorer

📊 Cost explorer

Detect

🔔 Alerting

🔧 Error reporting

🕒 Uptime checks

🔄 Synthetic monitoring

📊 SLOs

🔗 Observability Scopes

📄 Release Notes

<I

← Cloud Monitoring LAMP Qwik Start Dashboard

+ Add widget

📄 Annotations (2)

Group by

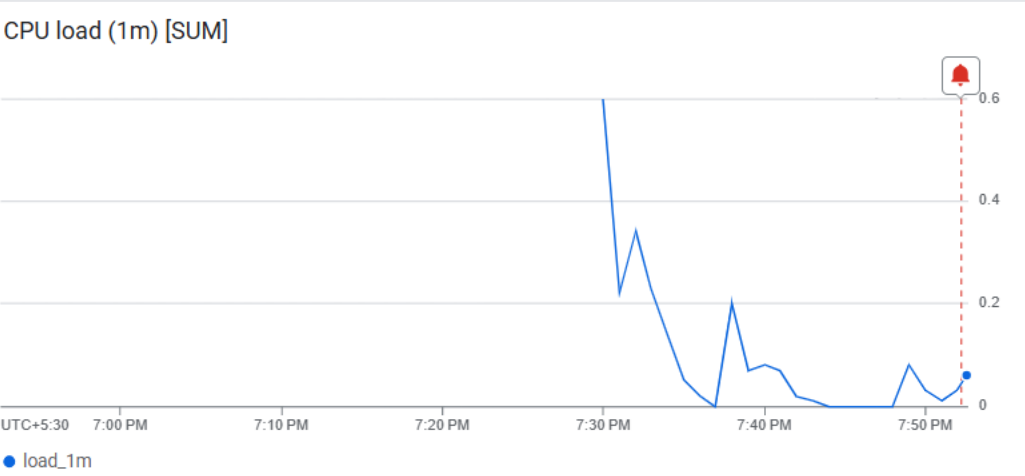
+ Filter

Saved 11/27/25, 7:56 PM

🟢 Autosave

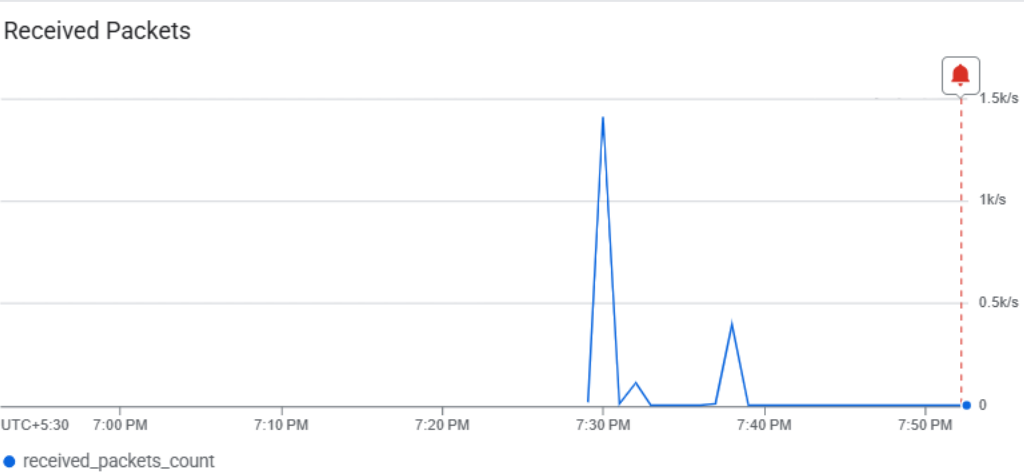
🔗 Share

CPU load (1m) [SUM]



Time	load_1m
7:30 PM	0.55
7:31 PM	0.25
7:32 PM	0.35
7:33 PM	0.25
7:34 PM	0.15
7:35 PM	0.10
7:36 PM	0.05
7:37 PM	0.05
7:38 PM	0.15
7:39 PM	0.10
7:40 PM	0.10
7:41 PM	0.05
7:42 PM	0.05
7:43 PM	0.05
7:44 PM	0.05
7:45 PM	0.05
7:46 PM	0.05
7:47 PM	0.05
7:48 PM	0.05
7:49 PM	0.05
7:50 PM	0.05

Received Packets



Time	received_packets_count
7:30 PM	1.4k/s
7:31 PM	0.1k/s
7:32 PM	0.1k/s
7:33 PM	0.1k/s
7:34 PM	0.1k/s
7:35 PM	0.1k/s
7:36 PM	0.1k/s
7:37 PM	0.1k/s
7:38 PM	0.1k/s
7:39 PM	0.1k/s
7:40 PM	0.4k/s
7:41 PM	0.1k/s
7:42 PM	0.1k/s
7:43 PM	0.1k/s
7:44 PM	0.1k/s
7:45 PM	0.1k/s
7:46 PM	0.1k/s
7:47 PM	0.1k/s
7:48 PM	0.1k/s
7:49 PM	0.1k/s
7:50 PM	0.1k/s

🔔 9+ Air: Poor Tomorrow

🏠 🔍 Search 🖱️ 📁 📄 📊 📈 📉 📋 📌 📍 📎 📏 📐 📑 📒 📓 📔 📕 📖 📗 📘 📙 📚 📛 📜 📝 📞 📟 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📽 📾 📿

ENG IN 📶 🔊 🔌 📶 07:56 PM 27-11-2025 🔔

Q

Dashboard

Catalog

Paths

Collections

Home > Develop Your Google Cloud Network > Cloud Monitoring: Qwik Start

ContentsCloud Monitoring: Qwik Start

Lab setup instructions and requirements

End Lab00:16:40

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Usernamestudent-02-44c3c12d53fdi

PasswordMEB8TZKXavPu

Project IDqwiklabs-gcp-01-0bae0d0i

Task 6. View your logs

Cloud Monitoring and Cloud Logging are closely integrated. Check out the logs for your lab.

1. Select **Navigation menu > Logging > Logs Explorer.**
2. Select the logs you want to see, in this case, you select the logs for the lamp-1-vm instance you created at the start of this lab:
 - Click on **All Resource.**
 - Select **VM Instance > lamp-1-vm** in the Resource drop-down menu.
 - Click **Apply.**

In the results section you can see the logs for your VM instance.

Check out what happens when you start and stop the VM instance.

To best see how Cloud Monitoring and Cloud Logging reflect VM instance changes,

100/100

Task 1. Create a Cloud Engine instance

Task 2. Add Apache2 HTTP Server to your instance

Task 3. Create an uptime check

Task 4. Create an alerting policy

Task 5. Create a dashboard and chart

Task 6. View your logs

Task 7. Check the uptime check results and triggered alerts

Congratulations!

< Previous

Next >

Observability Monitoring 📌

- Overview
- Dashboards**
- Application monitoring

Explore ^

- Metrics explorer
- Logs explorer**
- Log analytics
- Trace explorer
- Cost explorer

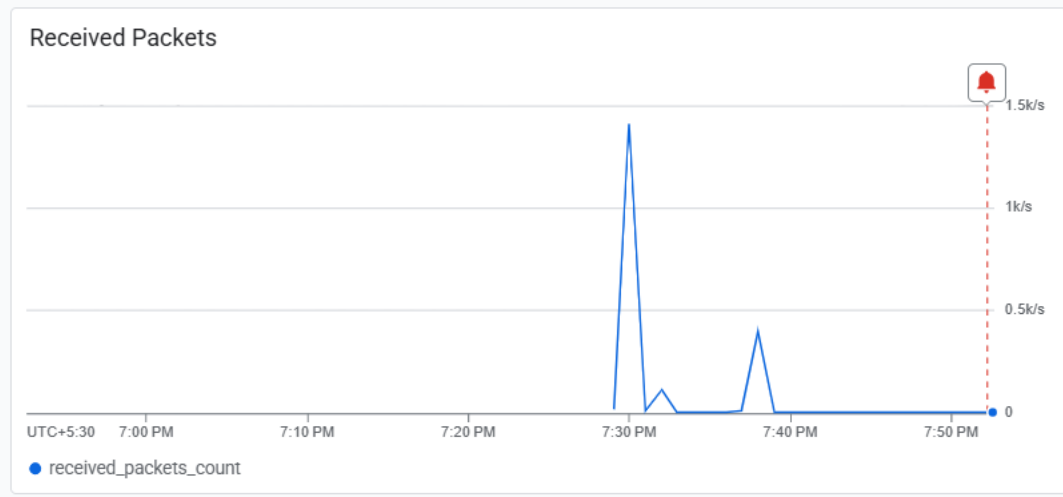
Detect ^

- Alerting
- Error reporting
- Uptime checks
- Synthetic monitoring
- SLOs

Observability Scopes >

- quwiklabs-gcp-01-0bae0d009b16

Release Notes



Run query

×

Apply

 $\geq$

54

OS Log

Dashboard

Catalog

Paths

Collections

Cloud Monitoring: Qwik Start

Contents

Cloud Monitoring: Qwik Start

Lab setup instructions and requirements

End Lab00:12:22

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username

student-02-44c3c12d53fdi

Password

MEB8TZKXavPu

Project ID

qwiklabs-gcp-01-0bae0d0f

makes it easier to view how changes to the VM are reflected in the logs

3. In the Compute Engine window, select the `lamp-1-vm` instance, click the three vertical dots at the right of the screen and then click **Stop**, and then confirm to stop the instance.

It takes a few minutes for the instance to stop.

4. Watch in the Logs View tab for when the VM is stopped.

5. In the VM instance details window, click the three vertical dots at the right of the screen and then click **Start/resume**, and then confirm. It will take a few minutes for the instance to re-start. Watch the log messages to monitor the start up.

Task 7. Check the uptime check results and triggered alerts

1. In the Cloud Logging window, select **Navigation menu > Monitoring > Uptime checks**. This view provides a list of all active uptime checks, and the status of each in different locations.

You will see Lamp Uptime Check listed. Since you have just restarted your

100/100

Task 1. Create a Compute Engine instance

Task 2. Add Apache2 HTTP Server to your instance

Task 3. Create an uptime check

Task 4. Create an alerting policy

Task 5. Create a dashboard and chart

Task 6. View your logs

Task 7. Check the uptime check results and triggered alerts

Congratulations!

Dashboard – qwiklabs-gcp-01-0ba...

Logs Explorer – Logging – qwiklabs-g

VM instances – Compute Engine

Apache2 Debian Default Page: It wor

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/compute/instances?project=qwiklabs-gcp-01-0bae0d009b16

🔍 | 🛡️

📅 | 📄 | 🗨️ 2 | ☰

Free AI Paraphrasin...

Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

🔍 Search (/) for resources, docs, products, and more

🔍 Search

✦ 📄 3 ? ⋮ 📌

🔧 Compute Engine

📄 Overview

🛡️ Security risk overview

Virtual machines

🗨️ VM instances

📄 Instance templates

👤 Sole-tenant nodes

📄 Machine images

🔧 TPUs

📊 Committed use discou...

📅 Reservations

🔧 Migrate to Virtual Mach...

Storage

📄 Disks

📄 Storage Pools

📄 Snapshots

🛒 Marketplace

📄 Release Notes

<|

VM instances

Create instance

Import VM

Refresh

Learn

Instances

Observability

Instance schedules

VM instances

Filter Enter property name or value

🔍

📄

☐	Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐	✔️	lamp-1-vm	us-east1-c			10.142.0.2 (nic0)	104.196.99.84 (nic0)	SSH ▾ ⋮

Related actions

🔧 Fix data protection gaps

New

Assess data protection gaps at no cost. You can use the Backup and DR Service to configure VM backups with backup vault storage.

📊 Monitor VMs

View outlier VMs across metrics like CPU and network

🔍 Explore VM logs

View, search, analyze, and download VM instance logs

🔧 Patch management

Schedule patch updates and view patch compliance on VM instances

⚙️ Load balance between VMs

Set up Load Balancing for your applications as your traffic and users grow

▶ Start / Resume

Stop

Suspend

🔧 Reset

Run maintenance

Configure data protection

Delete

Create a group based on this VM Preview

View network details

Create new machine image

View logs

View monitoring

Hide

📺 Trending videos

Stranger Things...

🪟 🔍 Search

👤 📁 🗨️ 📄 📅 📌

ENG IN 📶 🔊 🔌 08:00 PM 27-11-2025 📌

Dashboard – qwiklabs-gcp-01-0ba...

Logs Explorer – Logging – qwiklabs-g

VM instances – Compute Engine

Apache2 Debian Default Page: It wor

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/compute/instances?project=qwiklabs-gcp-01-0bae0d009b16

🔍 | 🛡️

📅 | 📄 | 🗨️ 2 | ☰

🔗 Free AI Paraphrasin...

📄 Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

🔗 qwiklabs-gcp-01-0bae0d009b16

🔍 Search (/) for resources, docs, products, and more

🔍 Search

🌟 📄 🔔 ⓘ ⋮ S

🔧 Compute Engine 🔔

📄 Overview

🛡️ Security risk overview

Virtual machines ^

🗨️ VM instances

📄 Instance templates

👤 Sole-tenant nodes

📄 Machine images

🔌 TPUs

📊 Committed use discou...

📅 Reservations

🔄 Migrate to Virtual Mach...

Storage ^

💾 Disks

💾 Storage Pools

📷 Snapshots

🛒 Marketplace

📄 Release Notes

<|

VM instances

Create instance Import VM Refresh

Learn

Instances Observability Instance schedules

VM instances

Filter Enter property name or value ⓘ ||

☐	Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect	
☐	🟡	lamp-1-vm	us-east1-c			10.142.0.2 (nic0)		SSH	⋮

Related actions

🔧 Fix data protection gaps
New
Assess data protection gaps at no cost. You can use the Backup and DR Service to configure VM backups with backup vault storage.

📊 Monitor VMs
View outlier VMs across metrics like CPU and network

📄 Explore VM logs
View, search, analyze, and download VM instance logs

🔌 Set up firewall rules
Control traffic to and from a VM instance

🔄 Patch management
Schedule patch updates and view patch compliance on VM instances

🔄 Load balance between VMs
🔗
Set up Load Balancing for your applications as your traffic and users grow

▶ Start / Resume

Stop

Suspend

🔄 Reset

Run maintenance

Configure data protection

Delete

Create a group based on this VM Preview

View network details

Create new machine image

View logs

View monitoring

VM instance stopped

🌤️ 9+ 15°C Clear

🪟 🔍 Search 🖱️ 📁 🗨️ 🌐 🛒 🛡️ 📄

⬆️ ENG IN 🔊 🔌 📶 08:02 PM 27-11-2025 🔔

Dashboard – qwiklabs-gcp-01-0ba...

Uptime checks – Monitoring – qv

VM instances – Compute Engine – qv

+

< > ↺ 🏠

🔖

🔗 https://console.cloud.google.com/monitoring/uptime?project=qwiklabs-gcp-01-0bae0d009b16

🔍 | 🛡️

📅 | 📄 | 🗨️ 2 | ☰

📄 Free AI Paraphrasin...

📄 Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

🔍 Search (/) for resources, docs, products, and more

🔍 Search

🌟 📄 1 ? ⋮ 📄

📊 Observability Monitoring

🏠 Overview

📊 Dashboards

📈 Application monitoring

Explore

📊 Metrics explorer

📋 Logs explorer

🔍 Log analytics

🔍 Trace explorer

📊 Cost explorer

Detect

🔔 Alerting

🔧 Error reporting

📺 Uptime checks

🔗 Synthetic monitoring

🕒 SLOs

🔗 Observability Scopes

📄 Release Notes

📄 Uptime checks

📄 + Create uptime check

+ 1 recommended alert

📖 Learn

🌱

Try the new Synthetic monitoring capability! New

Proactively monitor your application's critical user journeys by simulating user behavior. Detect and alert on failures, and receive detailed reports for debugging. [Learn more.](#)

Create Synthetic Monitor

View Documentation

Filter

Enter property name or value

Display Name	Asia Pacific	Europe	North America	South America	Policies
Lamp Uptime Check	✅	✅	⚠️	❌	1

https://console.cloud.google.com/monitoring/alerting?project=qwiklabs-gcp-01-0bae0d009...

9+ 15°C Clear

🪟 🔍 Search 🖱️ 📁 🗨️ 📄 📊 📄 📄

ENG IN 📶 🔊 🔌

08:04 PM 27-11-2025 🛎️

Dashboard – qwiklabs-gcp-01-0ba...

Alerting – Monitoring – qwiklabs

VM instances – Compute Engine – qv

+

< > ↺ 🏠

🔖

🔗

https://console.cloud.google.com/monitoring/alerting?project=qwiklabs-gcp-01-0bae0d009b16

🔍 | 🛡️

📅 | 📄 | 🗨️ 2 | ☰

📄 Free AI Paraphrasin...

📄 Transcript of Case St...

📁 GCP-LAB

📁 Hydra

📁 datawarehouse

☰ Google Cloud

👤 qwiklabs-gcp-01-0bae0d009b16

🔍 Search (/) for resources, docs, products, and more

🔍 Search

✦ 📺 1 ? ⋮ 📄

📊 Observability Monitoring

🏠 Overview

📊 Dashboards

📈 Application monitoring

Explore

📊 Metrics explorer

📋 Logs explorer

🔍 Log analytics

📊 Trace explorer

📊 Cost explorer

Detect

🔔 Alerting

🔔 Error reporting

🕒 Uptime checks

🔄 Synthetic monitoring

🕒 SLOs

🔗 Observability Scopes

📄 Release Notes

<|

Alerting

+ Create policy

✎ Edit notification channels

📖 Learn

🔔 Monitoring now supports both user-scoped and device-scoped Cloud Console Mobile notification channels

📄 Manage channels

📖 Learn more

🗑 Dismiss

Summary

Incidents firing

3 🚨

Incidents acknowledged

0

Alert policies

2

🔗 View all

Incidents

🔗 Show closed incidents

State	Severity	Policy name	Incident summary	Opened	
🚨	No severity	Lamp Uptime Check upti...	An uptime check on qwiklabs-gcp-01-0bae0d009b16 lamp-1-vm is failing.	Nov 27, 2025, 8:04:00 PM	▼
🚨	No severity	Inbound Traffic Alert	Network traffic for qwiklabs-gcp-01-0bae0d009b16 lamp-1-vm with metric labels {device=ens4, direction=rx} is a...	Nov 27, 2025, 8:02:48 PM	▼
🚨	No severity	Inbound Traffic Alert	Network traffic for qwiklabs-gcp-01-0bae0d009b16 lamp-1-vm with metric labels {device=ens4, direction=tx} is a...	Nov 27, 2025, 7:52:14 PM	▼

➔ See all incidents

Snoozes ? Create snooze

🔛 Show past snoozes

State	Name	Start time	End time
No rows to display			

➔ See all snoozes

🌤 9+ 15°C Clear

🪟 🔍 Search

👤 📁 🗨️ 🌐 📊 🛡️ 📄

⬆️ ENG IN 📶 🔊 🔌 08:05 PM 27-11-2025 📄